SCBNet: A Safety-Driven Class-adaptive Boundary-aware Network for Semantic Segmentation in Foggy Autonomous Driving Scenes

Yiting Zhang[1], Zhixiang Xu[1], Xiaoya Fan[1] and Zhong Wang[2]

***Abstract*— Semantic segmentation in autonomous driving degrades under fog, posing safety risks due to blurred boundaries of critical objects like pedestrians and vehicles. Existing methods often treat boundaries uniformly, neglecting semantic importance of critical objects, while failing to recover fog-degraded features and maintain structural alignment. To address these limitations, we propose SCBNet, a safety-driven framework that enhances robustness. Specifically, SCBNet integrates a Parallel Attention Enhancement Module to restore feature details, combined with a Class-Adaptive Edge Loss and a Boundary Consistency Loss to dynamically prioritize high-risk boundaries and ensure output coherence. Experiments on Foggy Cityscapes show SCBNet achieves 78.35% mIoU, outperforming state-of-the-art methods. Furthermore, our model exhibits superior zero-shot generalization on real-world foggy and adverse weather benchmarks, validating its practical effectiveness for safety-critical systems.**

***Index Terms*— Semantic Segmentation, Foggy Scene, Boundary-Aware Segmentation**

I. Introduction

Accurate perception is critical for safe autonomous driving, but foggy conditions degrade the performance of vision-based perception systems. Specifically, semantic segmentation struggles in fog due to low contrast and blurred boundaries. While state-of-the-art architectures, ranging from Mask2Former [1] to recent large-kernel and state-space models like InternImage [2] and VMamba [3], achieve remarkable performance in clear weather, their precision on critical object boundaries degrades dramatically under foggy conditions. This degradation poses a severe safety risk, as even minor boundary errors can lead to catastrophic decision-making failures, such as missing a pedestrian in a foggy scene, as shown in Fig. 1 (a).

Prior research has attempted to address this challenge through synthetic-to-real adaptation [4] or boundary-aware loss functions [5]. Even recent state-of-the-art paradigms, such as Masked Image Consistency (MIC) [6] and Vision-Language adaptation [7], which leverage context modeling and semantic priors to combat domain shifts, remain inadequate for autonomous driving. This is because they typically assign uniform weights to all boundaries regardless of their semantic importance. Our analysis shows safety-critical boundaries degrade twice as severely as background edges in fog, yet receive equal optimization priority. Consequently, safety-critical boundaries are not prioritized over non-critical ones. Furthermore, existing models often lack specific feature enhancement mechanisms to recover high-frequency details suppressed by fog.

To address these limitations, as shown in Fig. 1 (b), we propose SCBNet, an efficient framework that improves semantic segmentation in foggy scenes without requiring modifications to the backbone network. Extensive experiments on Foggy Cityscapes demonstrate that SCBNet achieves state-of-the-art performance, particularly in boundary accuracy.

In summary, the key contributions of this paper are:

- **Parallel Attention Enhancement Module (PAEM):** A decode-head module that employs parallel channel and spatial attention pathways to circumvent information bottlenecks in sequential processing, effectively recovering high-frequency details suppressed by fog.
- **Safety-Driven Boundary Supervision:** A dual-loss strategy comprising Class-Adaptive Edge Loss (L_{CAE}) to prioritize safety-critical boundaries and Boundary Consistency Loss (L_{cons}) to enforce geometric harmony between edge and segmentation predictions.
- **Efficient and Generalizable Design:** Built on a lightweight ResNet-50 backbone, SCBNet achieves state-of-the-art boundary accuracy while exhibiting strong zero-shot robustness across rain, snow, and night conditions.

II. Related Work

Semantic segmentation has advanced, incorporating sophisticated architectural innovations to improve scene understanding; however, despite these efforts, state-of-the-art methods remain fragile in adverse conditions like fog. Standard CNN-based approaches established a baseline by utilizing encoder-decoder designs to recover spatial details and object boundaries lost during downsampling. Simultaneously, capturing multi-scale context became a priority for distinguishing objects of varying sizes. This led to the development of dilated convolutions in the DeepLab family [8] and pyramid pooling in PSPNet [9], while HRNet [10] maintained high-resolution representations throughout the network. Recent works like InternImage [2] have further explored large-kernel convolutions to scale up effective receptive fields. More recently, the paradigm has shifted towards Vision Transformers. Pioneered by SegFormer [11], models like Mask2Former [1] and Mask DINO [12] reframe segmentation as mask classification, leveraging self-attention for long-range dependencies beyond the reach of local convolutions. Following these, the Segment Anything Model (SAM) [13]

*This work was not supported by any organization

[1]Yiting Zhang, Zhixiang Xu and Xiaoya Fan are with Dalian University of Technology, Dalian, China. {2669619340, 20223281063, xiaoyafan}@dlut.edu.cn

[2]Zhong Wang is with the Beijing Institute of Mathematical Sciences and Applications, Beijing, China. wangzhong@bimsa.cn

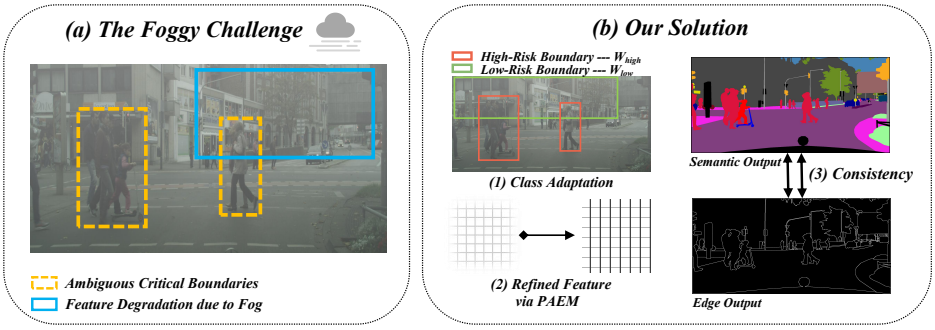


Fig. 1. An illustration of our proposed conceptual framework for robust semantic segmentation in foggy conditions. **(a) The Challenge:** Standard methods suffer from ambiguous boundaries for safety-critical objects and feature degradation. **(b) Our Solution:** We propose a SCBNet that addresses these challenges through three core innovations: (1) an Adaptive Edge Head supervised by our proposed Class-Adaptive Edge Loss (L_{CAE}), differentiating high- and low-risk boundaries, (2) a Parallel Attention Enhancement Module (PAEM) for feature enhancement, and (3) a Boundary Consistency Loss ($L_{\text{consistency}}$) to ensure synergy between the segmentation and edge outputs.

marked a shift toward foundation models. Most recently, VMamba [3] has explored state space models to achieve linear complexity with global modeling capabilities. However, despite these innovations, state-of-the-art methods are predominantly benchmarked on clear-weather datasets like Cityscapes [14]. Their reliance on clean visual cues makes them fragile in adverse conditions like fog, highlighting the need for robust, weather-resilient adaptation strategies.

With the increasing deployment of autonomous vehicles in real-world environments, the research focus has expanded from ideal clear-weather scenarios to robust perception in adverse conditions, particularly foggy scenes. Initial attempts typically adopted a multi-stage pipeline, applying image dehazing algorithms (e.g., FFA-Net [15] and deep priors [16]) before segmentation. However, these methods are computationally intensive and prone to error propagation when dehazing artifacts occur. Subsequently, end-to-end networks such as FIFO [17], as well as advanced Transformer-based architectures, were proposed to directly map foggy inputs to semantic masks. Domain adaptation methods [4] and recent consistency-based approaches such as MIC [6] further improve robustness, yet apply uniform optimization across the entire image. Consequently, they often fail to preserve fine-grained details on safety-critical edges. To address boundary degradation, general loss functions like Focal Loss [18] have been adapted. However, a key limitation persists: they do not explicitly prioritize boundaries based on the safety relevance of the adjacent classes, treating a pedestrian-road boundary the same as a sky-building boundary. Similarly, while attention mechanisms like CBAM [19] and Coordinate Attention [20] enhance features in clear images, their standard sequential integration often creates information bottlenecks when processing degraded foggy representations, limiting their efficacy in feature restoration.

III. Methodology

A. Overview

As illustrated in Fig. 2, SCBNet is built upon the DeepLabV3+ architecture. The high-level features extracted by the encoder and ASPP module are fed into our Parallel Attention Enhancement Module (PAEM), which refines the fog-degraded representations. The refined features then diverge into three parallel branches: a Main Segmentation Head for the final dense prediction, an Auxiliary Segmentation Head to facilitate deep supervision, and a dedicated Edge Head. The Edge Head is specifically supervised by our Safety-Driven Boundary Supervision mechanism to enforce geometric integrity.

B. Parallel Attention Enhancement Module

To enhance robustness to degraded features in foggy scenes, we embed a PAEM in the decoder's upsampling stage. Standard convolutional block attention module [19] connects its modules (CAM and SAM) sequentially, which may cause information loss. In foggy conditions where features are already degraded, such sequential processing creates a risky dependency chain: if the preceding module (e.g. Channel Attention) erroneously suppresses weak but critical cues due to noise, this error propagates and amplifies in the subsequent Spatial Attention step, rendering recovery impossible. Our PAEM design, shown in Fig. 2 (a), provides a solution to this problem.

Our PAEM module computes the CAM and SAM concurrently. For the input feature map $F \in \mathbb{R}^{C\times H\times W}$, the CAM aggregates spatial information by applying global average pooling and global max pooling across the channel dimension. The results are then fed into a shared multi-layer perceptron to generate the channel attention weights $M_c(F)$:

$$M_c(F) = \sigma\left(M_{\text{MLP}}(P_{\text{Avg}}(F)) + M_{\text{MLP}}(P_{\text{Max}}(F))\right), \quad (1)$$

where σ is Sigmoid, M_{MLP} is shared MLP, P_{Avg} is global average pooling, and P_{Max} is global max pooling.

The SAM compresses the channel dimension to generate two 2D feature maps representing global statistics across channels. These are then concatenated and fed into a 7×7 convolutional layer to generate the spatial attention weights $M_s(F)$:

$$M_s(F) = \sigma\left(f^{7\times 7}([\text{AvgPool}(F); \text{MaxPool}(F)])\right), \quad (2)$$

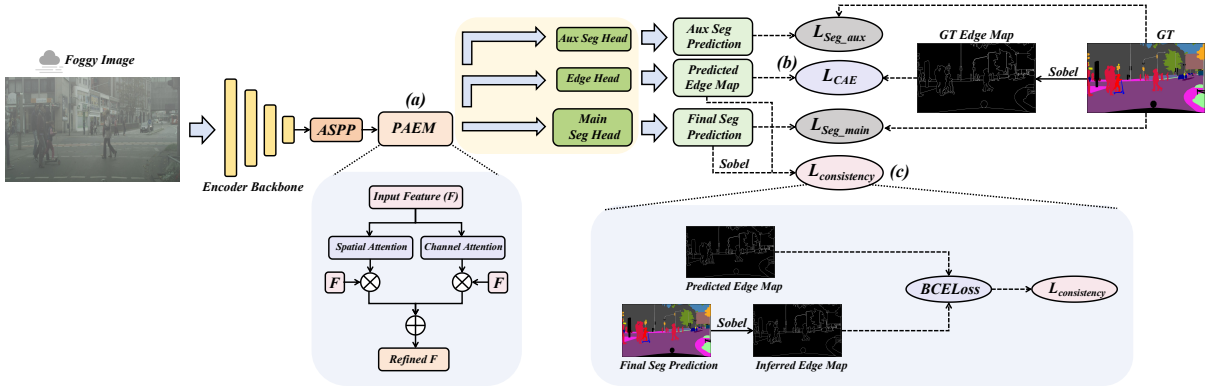


Fig. 2. An overview of our proposed SCBNet. The model is built upon the DeepLabV3+ architecture and incorporates three key innovations: (a) A Parallel Attention Enhancement Module (PAEM) is embedded in the decoder to refine features. (b) An Edge Head is supervised by a Class-Adaptive Edge Loss (L_{CAE}) which prioritizes critical object boundaries. (c) A Boundary Consistency Loss ($L_{consistency}$) is introduced to enforce agreement between the segmentation and edge prediction outputs, creating a feedback loop from the main segmentation head to the edge head.

where $f^{7\times7}$ represents a convolution operation with a 7×7 filter size, and $[\ ;\ ]$ denotes concatenation; AvgPool() and MaxPool() represent average-pooling and max-pooling operations performed along the channel axis.

Finally, the generated attention weights are multiplied element-wise with the original feature map F to refine the features. To adaptively balance the contributions of channel and spatial attention, two learnable parameters, α and β, are introduced. The weighted features are then summed and passed through a 1×1 convolutional layer $\mathcal{F}_{conv}$ to integrate the information and generate the enhanced feature map F':

$$F' = \mathcal{F}_{conv}\left(\alpha \cdot (M_c(F) \otimes F) + \beta \cdot (M_s(F) \otimes F)\right), \quad (3)$$

where $\otimes$ denotes element-wise multiplication, $\mathcal{F}_{conv}$ represents a 1×1 convolution operation for feature fusion, and α and β are learnable scalar weights that adaptively balance the attention branches.

C. Safety-Driven Boundary Supervision

1) ***Class-Adaptive Edge Loss****:* Traditional boundary losses treat all boundaries equally, ignoring semantic importance (e.g., a pedestrian-road boundary is more critical than sky-building). We propose the Class-Adaptive Edge Loss (L_{CAE}) to address this, applying it to a dedicated auxiliary edge head (Fig. 2 (b)). L_{CAE} adjusts edge loss weights based on boundary class safety relevance.

First, we assign a safety weight W_c to each class c, giving higher weights to critical classes (e.g. pedestrians, roads etc.) and lower weights to background classes, as detailed in Table I. Notably, W_c can be manually adjusted for specific deployment scenarios—for instance, increasing W_{person} when pedestrians are frequently blurred or occluded, enabling flexible model adaptation without retraining.

Next, we extract a pixel-level ground truth boundary mask, M_{edge}, by applying a Sobel operator to the ground truth segmentation map. For each pixel i identified as a boundary pixel ($i \in M_{\text{edge}}$), it separates two adjacent regions of different classes, denoted as c_1 and c_2. The weight for this specific boundary pixel, $W_{\text{boundary},i}$, is calculated as the product of the safety weights of the two classes it separates:

$$W_{\text{boundary},i} = W_{c_1} \times W_{c_2}. \quad (4)$$

We use the product to ensure that boundaries between two critical classes (e.g., person-road) receive higher weights than those involving at least one non-critical class (e.g., person-sky), as both classes contribute to the boundary's safety relevance. For corner pixels where multiple classes meet, we take the maximum weight among all pairwise products. These corner cases represent a negligible portion of boundary pixels and have minimal impact on training. For non-boundary pixels, the weight is set to a default value of 0. This method transforms the weights of individual classes into weights for the boundaries between them, effectively prioritizing more critical edges over less critical ones.

The L_{CAE} is formulated as a weighted Focal Loss as follows:

$$L_{\text{CAE}} = -\frac{1}{N}\sum_{i=1}^{N} W_{\text{boundary},i}\Big(\alpha(1-p_i)^{\gamma} y_i \log(p_i) + (1-\alpha)p_i^{\gamma}(1-y_i)\log(1-p_i)\Big), \quad (5)$$

TABLE I
CLASS-ADAPTIVE SAFETY WEIGHTS (W_c) FOR ALL 19 CLASSES

Class	Weight	Class	Weight
person	1.0	traffic light	0.80
rider	1.0	traffic sign	0.80
car	0.92	pole	0.70
truck	0.92	wall	0.65
bus	0.92	fence	0.65
train	0.92	building	0.55
motorcycle	0.95	vegetation	0.50
bicycle	0.95	terrain	0.50
road	0.85	sky	0.45
sidewalk	0.85		

TABLE II

COMPREHENSIVE ABLATION STUDY OF OUR PROPOSED COMPONENTS ON THE FOGGY CITYSCAPES VALIDATION SET

ID	Model Configuration	Baseline	L_{CAE}	PAEM	L_{cons}	mIoU(%)	mAcc(%)	aAcc(%)
1	Baseline	✓				54.97	66.11	92.59
2	Baseline + L_{CAE}	✓	✓			64.48	76.37	93.29
3	Baseline + PAEM	✓		✓		74.50	83.19	95.40
4	Baseline + L_{CAE} + PAEM	✓	✓	✓		74.80	83.21	95.42
5	Baseline + L_{CAE} + L_{cons}	✓	✓		✓	77.69	83.60	95.99
6	**Baseline + L_{CAE} + PAEM + L_{cons}**	✓	✓	✓	✓	**78.35**	**86.16**	**96.17**

where N is the total pixels, $y_i \in \{0, 1\}$ is the ground truth (1 for edge), p_i is the predicted edge probability, and α, γ are the Focal Loss focusing parameters.

By incorporating $W_{\text{boundary},i}$, our L_{CAE} directs the Edge Head—rather than the segmentation heads—to prioritize learning semantically important boundaries, isolating boundary-sharpness learning from standard per-pixel classification.

*2) **Boundary Consistency Loss**:* In SCBNet, we enforce consistency between the edge predictions from the dedicated edge head and the boundaries derived from the semantic segmentation head (Fig. 2 (c)). This ensures that edge detections geometrically align with class transitions, thereby refining segmentation masks.

This loss, calculated in two steps, bridges the gap between the multi-class segmentation and binary edge detection heads. First, we convert the semantic segmentation output: The multi-channel output is converted to a class map by selecting the class with the highest probability at each pixel. A Sobel operator is applied to this map to generate a binary edge map, $E_{\text{seg}} \in \{0, 1\}^{H \times W}$, which serves as the consistency target. Next, for the Boundary Consistency Loss calculation, the edge detection head's output, $E_{\text{edge}} \in [0, 1]^{H \times W}$, is a probability map. We enforce consistency by minimizing the difference between E_{edge} and E_{seg} using a Binary Cross-Entropy (BCE) loss:

$$L_{\text{consistency}} = \text{BCE}(E_{\text{seg}}, E_{\text{edge}}). \tag{6}$$

The edge detection head is supervised by combining L_{CAE} and $L_{\text{consistency}}$:

$$L_{\text{edge_head}} = \lambda_{\text{edge}} L_{\text{CAE}} + \lambda_{\text{cons}} L_{\text{consistency}}, \tag{7}$$

where λ_{edge} and λ_{cons} are hyperparameters that balance the two supervisory signals.

By minimizing this loss, we compel the Edge Head's prediction to track the segmentation head's evolving output rather than only the static Sobel-derived ground truth, forming a feedback bridge between the two branches.

*3) **Total Loss Function**:* The entire model is trained end-to-end by jointly optimizing three complementary objectives: $L_{\text{seg_main}}$ drives the Main Segmentation Head's final prediction and is the only term retained at inference; $L_{\text{seg_aux}}$ applies deep supervision to an earlier decoder stage, easing gradient flow through PAEM; and $L_{\text{edge_head}}$ injects boundary-specific supervision that per-pixel Cross-Entropy cannot provide. The latter two are discarded after training and serve purely as regularizers shaping the shared backbone features used by the Main Head.

The final loss function is formulated as:

$$L_{\text{total}} = w_{\text{main}} L_{\text{seg_main}} + w_{\text{aux}} L_{\text{seg_aux}} + w_{\text{edge}} L_{\text{edge_head}}, \tag{8}$$

where $L_{\text{seg_main}}$ and $L_{\text{seg_aux}}$ are standard Cross-Entropy segmentation losses. $L_{\text{edge_head}}$ is the combined edge loss (Section 3.3), and w_{main}, w_{aux}, w_{edge} are balancing weights.

IV. EXPERIMENTS

A. Experimental Setup

1) Dataset: We train on **Foggy Cityscapes** [4], a synthetic foggy dataset derived from **Cityscapes** [14] with 19 semantic classes. Training exclusively on synthetic fog allows us to evaluate generalization to real-world adverse conditions without domain adaptation. For testing, we evaluate on multiple unseen real-world benchmarks without re-training. For Sim2Real fog generalization, we use **Foggy Zurich (FZ)** [26] (real foggy urban scenes), **Foggy Driving (FD)** [4] (real highway fog), and **ACDC-Fog** [27] (dense fog scenes). For cross-weather generalization, we test on **ACDC-Rain**, **ACDC-Snow**, and **ACDC-Night** [27], which cover rainy, snowy, and nighttime driving scenarios. All test datasets share the same 19 Cityscapes classes, enabling direct evaluation of generalization across diverse adverse conditions.

2) Implementation Details: All experiments used 2 NVIDIA RTX 3090 GPUs (batch size 1/GPU), trained for 80k iterations with AdamW (lr 6×10^{-3}, weight decay 0.01, polynomial schedule, power 0.9). Input images were resized to 1024×512 and augmented with random horizontal flipping, random scaling (0.5-2.0), and color jittering. The loss weights were set to $\lambda_{\text{seg}} = 1.0$, $\lambda_{\text{edge}} = 1.0$, and $\lambda_{\text{cons}} = 0.5$. For the Focal Loss parameters in L_{CAE}, we used $\alpha = 0.25$ and $\gamma = 2.0$.

3) Evaluation Metrics: We report mean Intersection over Union (mIoU) as our primary metric, alongside Mean Accuracy (mAcc) and Overall Accuracy (aAcc) as auxiliary metrics.

B. Ablation Study

We performed ablation studies to demonstrate the effectiveness of our components which is showed in Table II.

Baseline Model Performance: Our baseline model, DeepLabV3+, established the initial performance benchmark

TABLE III

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE FOGGY CITYSCAPES VALIDATION SET

ID	Method	Backbone	mIoU (%)	mAcc (%)	aAcc (%)
1	DANet [21]	ResNet-50	73.70	81.65	95.50
2	CCNet [22]	ResNet-50	73.87	82.51	95.61
3	HANet [23]	ResNet-50	74.17	82.18	95.90
4	SegFormer [11]	-	76.10	83.40	95.75
5	Mask2Former [1]	ResNet-50	76.45	84.68	95.94
6	Mask DINO [12]	ResNet-50	77.21	85.53	96.06
7	VMamba [3]	-	77.51	85.13	95.66
8	GCNet-S [24]	-	77.30	85.11	95.34
9	SegMAN [25]	-	77.82	85.11	95.93
10	**SCBNet (Ours)**	**ResNet-50**	**78.35**	**86.16**	**96.17**

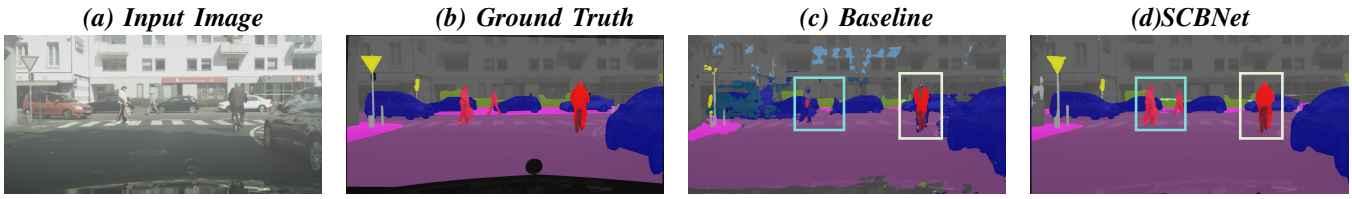


Fig. 3. Qualitative comparison on the Foggy Cityscapes validation set. From left to right: (a) Input foggy image, (b) Ground Truth, (c) Baseline (DeepLabV3+) prediction, and (d) SCBNet prediction. Key areas are highlighted with blue and white boxes for detailed comparison.

on the Foggy Cityscapes validation set (ID 1). Its results reflected the challenge posed by foggy conditions to a standard segmentation model, providing a clear reference point to quantify the improvements from our proposed components.

Individual Component Analysis: As shown in Table II, adding L_{CAE} improves mIoU by 9.51%, validating the benefit of safety-aware weighting. Notably, PAEM yields the largest single-component gain, confirming that parallel processing recovers fog-degraded structural details effectively.

Synergistic Effects of Components: We examined component interactions to understand their combined efficacy. While adding L_{CAE} to PAEM (ID 4) yielded minimal gains, pairing L_{CAE} with L_{cons} (ID 5) achieves 77.69% mIoU, demonstrating strong synergy between boundary weighting and geometric consistency. The full model integrates all components for optimal performance, confirming that L_{cons} effectively bridges boundary supervision and feature enhancement to produce coherent predictions in foggy scenes.

Figure 3 shows a qualitative comparison: our method produces cleaner, more accurate segmentation for safety-critical objects (e.g., cyclists, pedestrians) than the baseline, particularly under dense fog.

C. Comparison with State-of-the-Art

We compare our SCBNet with recent state-of-the-art semantic segmentation methods on the Foggy Cityscapes validation set, as shown in Table III. SCBNet achieves 78.35% mIoU, outperforming all baselines including VMamba and SegMAN. The superior performance is primarily attributed to our design explicitly addressing feature degradation in fog. Unlike general-purpose Transformer or Mamba architectures, which may overlook fine-grained edge information in low-contrast scenes, SCBNet successfully integrates global context with boundary-aware features, resulting in superior mAcc (86.16%) and aAcc (96.17%).

TABLE IV

GENERALIZATION PERFORMANCE (MIOU %) ON REAL-WORLD FOGGY DATASETS (FZ: FOGGY ZURICH, FD: FOGGY DRIVING, A-FOG: ACDC-FOG.)

Method	Testing Data		
	FZ	FD	A-Fog
DANet [21]	45.7	51.2	65.1
CCNet [22]	46.1	51.5	65.4
SegFormer [11]	48.9	55.8	67.9
Mask2Former [1]	50.6	56.1	68.7
Mask DINO [12]	50.5	56.5	68.9
VMamba [3]	49.8	55.9	68.4
GCNet-S [24]	49.7	56.7	68.4
SegMAN [25]	50.1	57.1	68.8
SCBNet (Ours)	**51.3**	**57.3**	**69.5**

D. Generalization Analysis on Unseen Domains

Beyond the primary benchmark, we evaluate SCBNet's generalization on two unseen scenarios: simulation-to-reality fog adaptation and cross-weather robustness. All models are trained on Foggy Cityscapes and evaluated directly without retraining or domain adaptation.

1) Simulation-to-reality: To evaluate practical utility, we compared SCBNet against state-of-the-art segmentation architectures, all trained on synthetic fog. As shown in Table IV, SCBNet outperforms both standard CNNs and advanced Transformer/Mamba models (e.g., Mask2Former, VMamba) on real-world benchmarks. This demonstrates that our safety-driven design with boundary-aware supervision effectively improves Sim2Real generalization on foggy scenes.

TABLE V

CROSS-WEATHER GENERALIZATION PERFORMANCE (MIOU %) ON THE ACDC VALIDATION SET (A-SNOW: ACDC-SNOW, A-RAIN: ACDC-RAIN, A-NIGHT: ACDC-NIGHT)

Method	Testing Data		
	A-Snow	A-Rain	A-Night
DANet [21]	44.1	45.8	43.1
CCNet [22]	44.8	46.1	43.4
SegFormer [11]	47.5	48.8	46.5
Mask2Former [1]	48.9	49.7	47.5
Mask DINO [12]	49.2	50.1	48.0
VMamba [3]	48.5	49.5	47.2
GCNet-S [24]	49.2	49.4	47.9
SegMAN [25]	49.5	49.7	48.2
SCBNet (Ours)	**50.4**	**51.3**	**49.4**

2) Generalization to Other Adverse Conditions: To evaluate cross-weather generalization, we directly test the model trained on Foggy Cityscapes (without retraining on other weather condition) on ACDC-Rain, Snow, and Night. As shown in Table V, SCBNet achieves the highest mIoU across all conditions, surpassing sophisticated competitors like Mask DINO and VMamba. This zero-shot cross-weather performance demonstrates that our boundary-aware design captures generalizable structural patterns beyond fog-specific features, maintaining robustness across diverse adverse conditions where visual details are similarly degraded.

V. CONCLUSION

In this paper, we presented SCBNet, which improves safety-aware boundary perception in foggy scenes through three synergistic components — L_{CAE}, PAEM, and $L_{consistency}$ — achieving state-of-the-art performance on Foggy Cityscapes without modifying the backbone.

Future work includes exploring dynamic learning of safety weights in CAELoss based on scene context, enabling user-adaptive weight adjustment for deployment-specific safety priorities, and extending our boundary-aware design to newer architectures such as Vision Transformers and diffusion-based models for enhanced generalization.

REFERENCES

[1] Bowen Cheng, Ishan Misra, Alexander G Schwing, et al., "Masked-attention mask transformer for universal image segmentation," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2022, pp. 1290–1299.

[2] Wenhai Wang, Jifeng Dai, Zhe Chen, Zhen Huang, Zhiqiang Li, Xizhou Zhu, Xiaowei Hu, Tong Lu, Lewei Lu, Hongsheng Li, et al., "Internimage: Exploring large-scale CNN based foundation models," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023.

[3] Yue Liu, Yunjie Tian, Yuzhong Zhao, Hongtian Yu, Lingxi Xie, Yaowei Wang, Qixiang Ye, Jianbin Jiao, and Yunfan Liu, "Vmamba: Visual state space model," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024.

[4] Christos Sakaridis, Dengxin Dai, and Luc Van Gool, "Semantic foggy scene understanding with synthetic data," *Int. J. Comput. Vis.*, vol. 126, pp. 973–992, 2018.

[5] Hoel Kervadec, Jihene Bouchouicha, Jose Dolz, et al., "Boundary loss for highly unbalanced segmentation," in *Int. Conf. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, 2019, pp. 284–292.

[6] Lukas Hoyer, Dengxin Dai, Haoran Wang, and Luc Van Gool, "MIC: Masked image consistency for context-enhanced domain adaptation," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023.

[7] Yongming Rao, Wenliang Zhao, Guangyi Chen, Yansong Tang, Zheng Zhu, Guan Huang, Jie Zhou, and Jiwen Lu, "DenseCLIP: Language-guided dense prediction with context-aware prompting," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2022.

[8] Liang-Chieh Chen, George Papandreou, Florian Schroff, and Hartwig Adam, "Rethinking atrous convolution for semantic image segmentation," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2017, pp. 6726–6735.

[9] Hengshuang Zhao, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia, "Pyramid scene parsing network," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2017, pp. 2881–2890.

[10] Jingdong Wang, Ke Sun, Tianheng Cheng, et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, 2020.

[11] Enze Xie, Wenhai Wang, Zhiding Yu, et al., "Segformer: Simple and efficient design for semantic segmentation with transformers," in *Adv. Neural Inform. Process. Syst.*, 2021, pp. 12077–12090.

[12] Feng Li, Hao Zhang, Huaizhe Wang, et al., "Mask dino: Towards a unified transformer-based framework for object detection and segmentation," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023, pp. 17441–17451.

[13] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollar, and Ross Girshick, "Segment anything," in *Int. Conf. Comput. Vis.*, October 2023, pp. 4015–4026.

[14] Marius Cordts, Mohamed Omran, Sebastian Ramos, et al., "The cityscapes dataset for semantic urban scene understanding," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2016, pp. 3213–3223.

[15] Xu Qin, Zhilin Wang, Yuanchao Bai, Xiaodong Xie, and Huizhu Jia, "FFA-Net: Feature fusion attention network for single image dehazing," in *AAAI*, 2020.

[16] Rui Wu, Tao Zheng, Zongyu Guo, and Xuanhong Zhang, "Ridcp: Revitalizing the dark channel prior for image dehazing," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023, pp. 17849–17858.

[17] Sohyun Lee, Taeyoung Son, and Suha Kwak, "Fifo: Learning fog-invariant features for foggy scene segmentation," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2022, pp. 18911–18921.

[18] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár, "Focal loss for dense object detection," in *Int. Conf. Comput. Vis.*, 2017, pp. 2980–2988.

[19] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon, "Cbam: Convolutional block attention module," in *Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.

[20] Qibin Hou, Daquan Zhou, and Jiashi Feng, "Coordinate attention for efficient mobile network design," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2021, pp. 13713–13722.

[21] Jun Fu, Jing Liu, Haijie Tian, et al., "Dual attention network for scene segmentation," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2019, pp. 3146–3154.

[22] Zilong Huang, Xinggang Wang, Lichao Huang, et al., "Ccnet: Criss-cross attention for semantic segmentation," in *Int. Conf. Comput. Vis.*, 2019, pp. 603–612.

[23] Sungha Choi, Joanne Kim, and Jaegul Choo, "Cars can't fly up in the sky: Improving urban-scene segmentation via height-driven attention networks," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2020, pp. 9373–9381.

[24] Guoyu Yang, Yuan Wang, Daming Shi, and Yanzhong Wang, "Golden cudgel network for real-time semantic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2025, pp. 25367–25376.

[25] Yunxiang Fu, Meng Lou, and Yizhou Yu, "Segman: Omni-scale context modeling with state space models and local attention for semantic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2025, pp. 19077–19087.

[26] Christos Sakaridis, Dengxin Dai, Simon Hecker, and Luc Van Gool, "Model adaptation with synthetic and real data for semantic dense foggy scene understanding," in *Eur. Conf. Comput. Vis.*, 2018, pp. 687–704.

[27] Christos Sakaridis, Haoran Wang, Ke Li, et al., "Acdc: The adverse conditions dataset with correspondences for robust semantic driving scene perception," in *Int. Conf. Comput. Vis.*, 2021, pp. 10765–10775.